POWER CALCULATIONS

TurtleBot Solenoid Integration

Solenoid Model: JF-0826B

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1. Overview

During testing, the solenoid was powered using a separate power source instead of the main TurtleBot power rail. The solenoid operates at approximately 12 V.

1.1 Batteries Used

Battery	Voltage / Capacity	Purpose
DC-168	12.6 V, 1800 mAh	Solenoid ONLY
3S LiPo (IDP)	11.1 V	All other systems

2. Component Power Summary

Component	Voltage (V)	Current (A)	Max Current (A)	Power (W)	Max Power (W)
Solenoid (normal runtime)	12	1.786	2.000	21.432	24.000
TurtleBot (boot)	11.1	0.889	No Surge	9.868	9.868
TurtleBot (standby)	11.1	0.544	No Surge	6.038	6.038
TurtleBot (teleop)	11.1	0.650	0.742	7.215	8.236

3. Solenoid Battery Analysis

3.1 Lab Power Supply Observations

- Supply voltage: ~12 V
- Typical running current (after transient): 1.786 A
- Short firing intervals: current fluctuates between ~1 A and 1.786 A
- Continuous ON: current rises to 2 A cap after ~15 seconds

The solenoid is not designed for continuous operation. It fires only in short pulses; sustained use causes significant heating.

3.2 Power Calculation

Formula: $P = V \times I$

Lower estimate (short-pulse transient):

$$P = 12.6\ V \times 1.4\ A = 17.64\ W$$

Typical measured running power:

$$P = 12.6\ V \times 1.786\ A = 22.50\ W$$

Worst-case (capped current):

$$P = 12.6\ V \times 2.0\ A = 25.20\ W$$

Solenoid power draw: 17.64 W to 22.5 W during short activations; worst-case continuous = 25.2 W.

3.3 Solenoid Battery Runtime (DC-168: 12.6 V, 1800 mAh)

Battery energy:

$$E = 12.6\ V \times 1.8\ Ah = 22.68\ Wh$$

Operating Condition	Current (A)	Power (W)	Runtime
Short-pulse (low transient)	1.000	12.60	108 min
Typical measured current	1.786	22.50	~60 min
Worst-case (capped, continuous)	2.000	25.20	54 min

Since the solenoid fires only in short pulses, average current draw is well below the continuous-ON values. Actual runtime will significantly exceed the estimates above.

Mission runtime is 25 minutes. The DC-168 battery provides adequate energy margin for solenoid operation under pulsed conditions.

4. TurtleBot Battery Endurance (3S LiPo)

Parameter	Value
TurtleBot battery energy (approx.)	19 Wh
Standby endurance	~2.5 hr
Max teleop endurance	~1.5 hr

5. Final Design Decision

A separate battery/power source was used for the JF-0826B solenoid actuator.

This was a major improvement over the PDR architecture (single TurtleBot battery) because it isolates the high-current switching transients from the TurtleBot power rail, eliminating the risk of:

- Voltage sag on the main rail during solenoid firing
- Brownout affecting the Raspberry Pi or OpenCR
- Unstable behaviour in control electronics

With the separate solenoid power source, the robot successfully completed the mission at both stations.

6. Notes & Important Observations

- If the solenoid is kept ON continuously, current slowly increases and reaches the 2 A lab supply limit after ~15 seconds.
- The solenoid is never intended to stay ON for extended periods during the mission — it is only fired in short pulses to prevent overheating.
- Separate power architecture was used to isolate the high-current solenoid load from the Raspberry Pi and OpenCR.

7. Assumptions

- Power consumption is assumed constant within each operating mode (boot / standby / teleop).
- Boot-up power is treated as a short transient and neglected in endurance estimates.
- Teleop runtime is estimated using the measured max teleop power (worst-case continuous draw).
- Solenoid is assumed to not be activated during TurtleBot navigation, or its duty cycle is negligible.
- Battery voltage and capacity are assumed constant and equal to rated values.
- Power conversion losses and internal resistance / voltage sag are neglected (first-order estimation).
- Temperature, battery ageing, and state-of-charge effects are neglected.